

CASE STUDY: MNIST DATASET

Practical PCA

Dimensionality Reduction
& Data Visualisation

Stack: Python / Scikit-Learn / Plotly

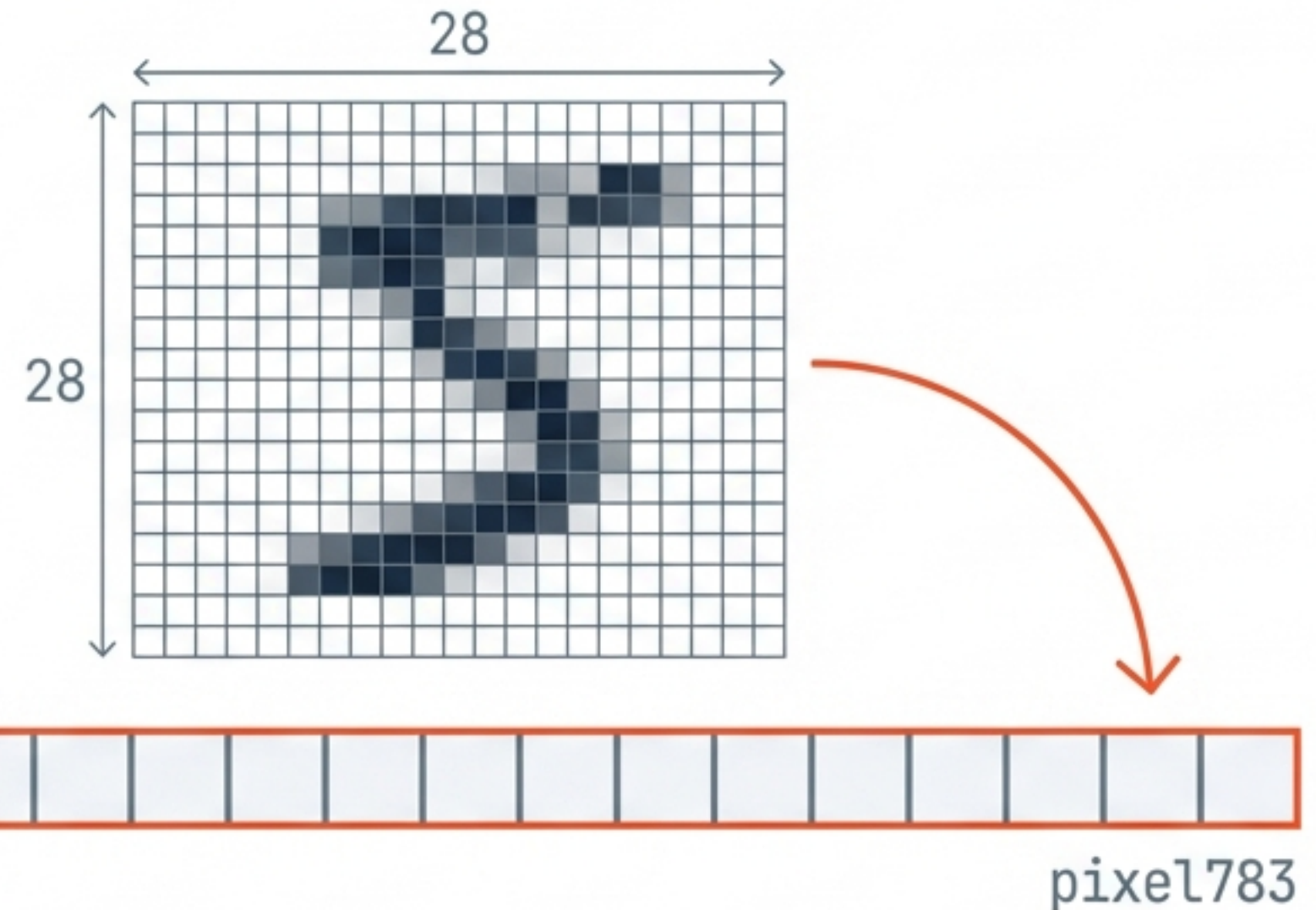


THE DATASET: 784 DIMENSIONS OF COMPLEXITY

Source: MNIST (42,000 handwritten digits).

Structure: 28x28 pixel grayscale grid.

The Challenge: To a machine, an image is not a shape; it is a **flattened vector** of 784 separate features. Every pixel is a dimension.



	label	pixel0	pixel1	pixel2	pixel3	...	pixel781	pixel782	pixel783
Row 0	1	0	0	0	0	...	0	0	0
Row 1	0	0	0	22	142	...	88	14	0

THE CONTROL EXPERIMENT: KNN ON RAW DATA

Establishing a performance baseline with K-Nearest Neighbours.

96.6%

Accuracy Score

The model effectively identifies digits using all 784 pixels.

High Latency

Computation Cost

Calculating Euclidean distance in 784-dimensional space requires comparing every point against 33,000 training samples. This is computationally expensive.

Predictive Power

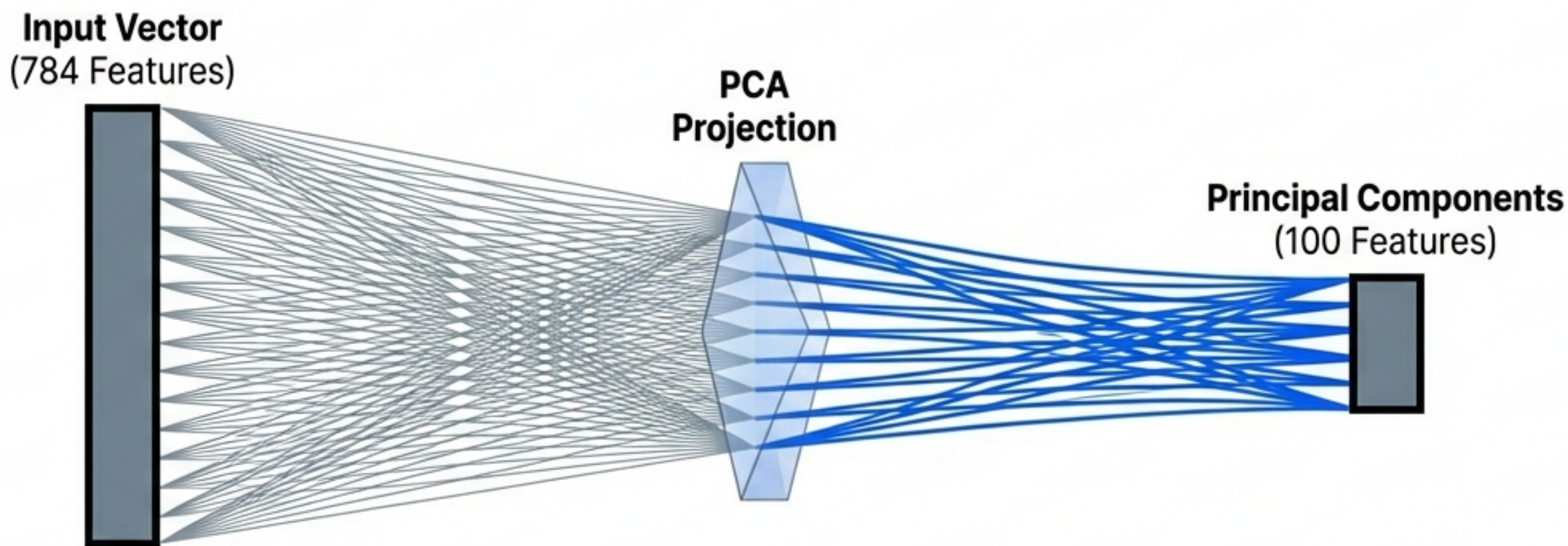


Training Efficiency



APPLYING PCA: COMPRESSING THE FEATURE SPACE

Methodology using Scikit-Learn to reduce dimensions while preserving variance.



```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

```
# 1. Standardise the data (Mean Centring)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_train)
```

```
# 2. Project to 100 Components
pca = PCA(n_components=100)
pca = PCA(n_components=0)
X_pca = pca.fit_transform(X_scaled)
```

Performance Analysis: Raw vs. Transformed

Baseline Model

Raw Pixel Data

784 Dimensions

96.6% Accuracy



Slow

PCA Transformed

100 Principal Components

100 Dimensions

95.5% Accuracy -1.1% Delta



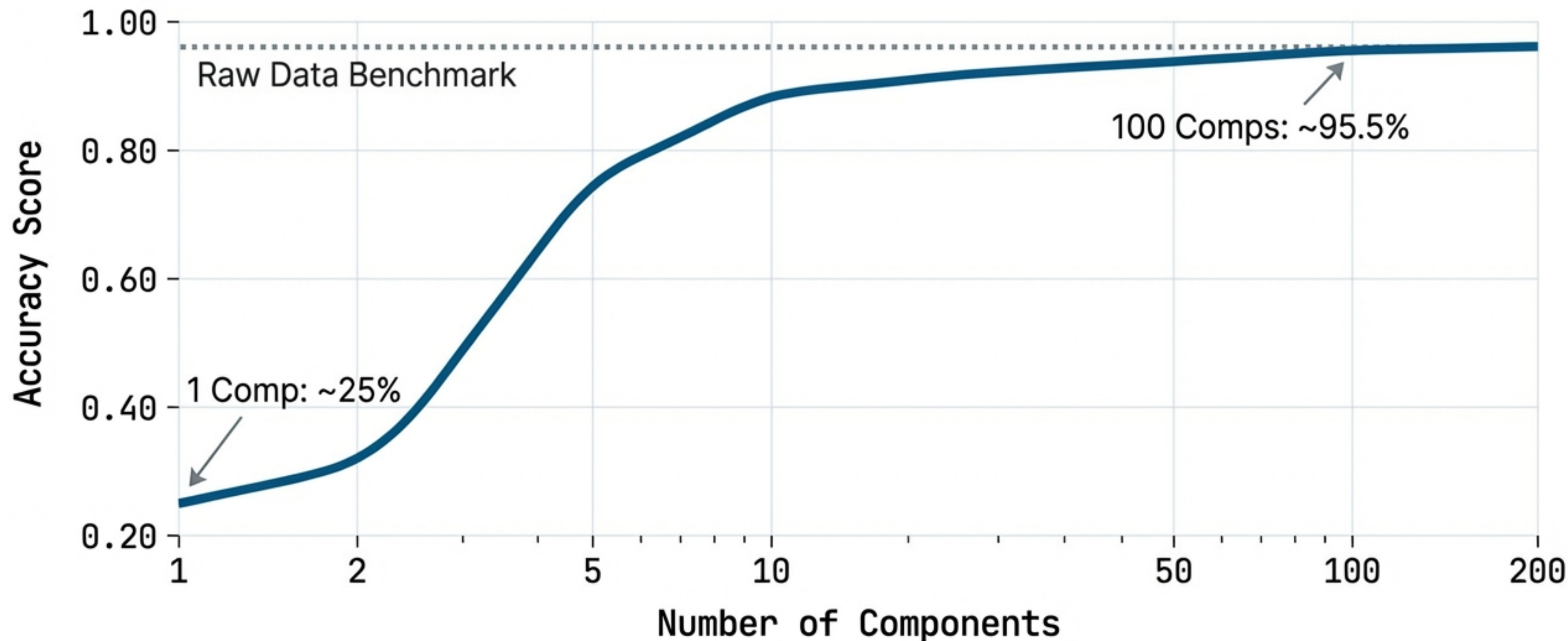
Fast

90% Data Reduction

Result: We retain nearly all predictive power while discarding the vast majority of the raw data volume.

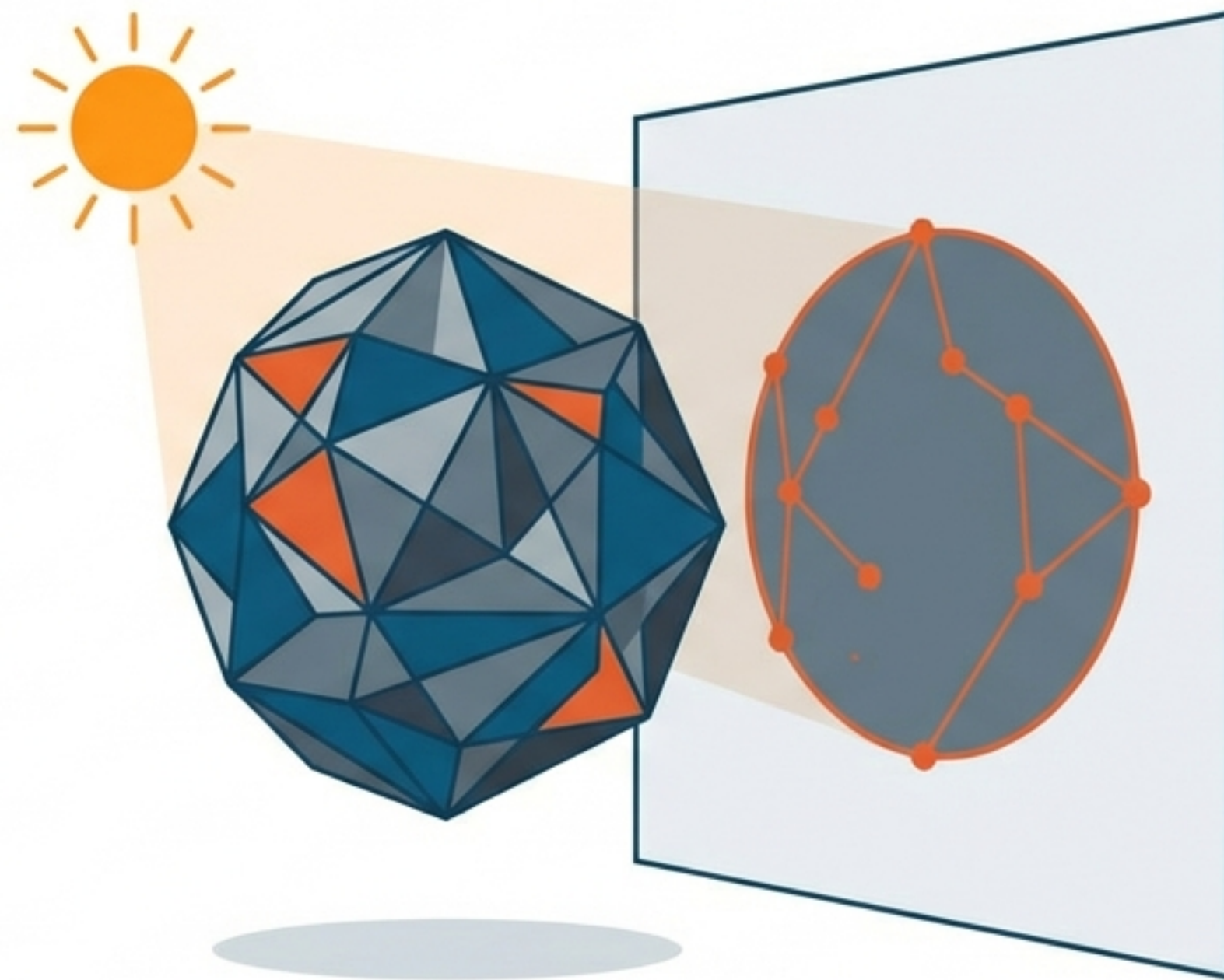
THE ACCURACY CURVE: ITERATIVE OPTIMISATION

How many components are needed before accuracy plateaus?



Use Case 2: Visualising the Invisible

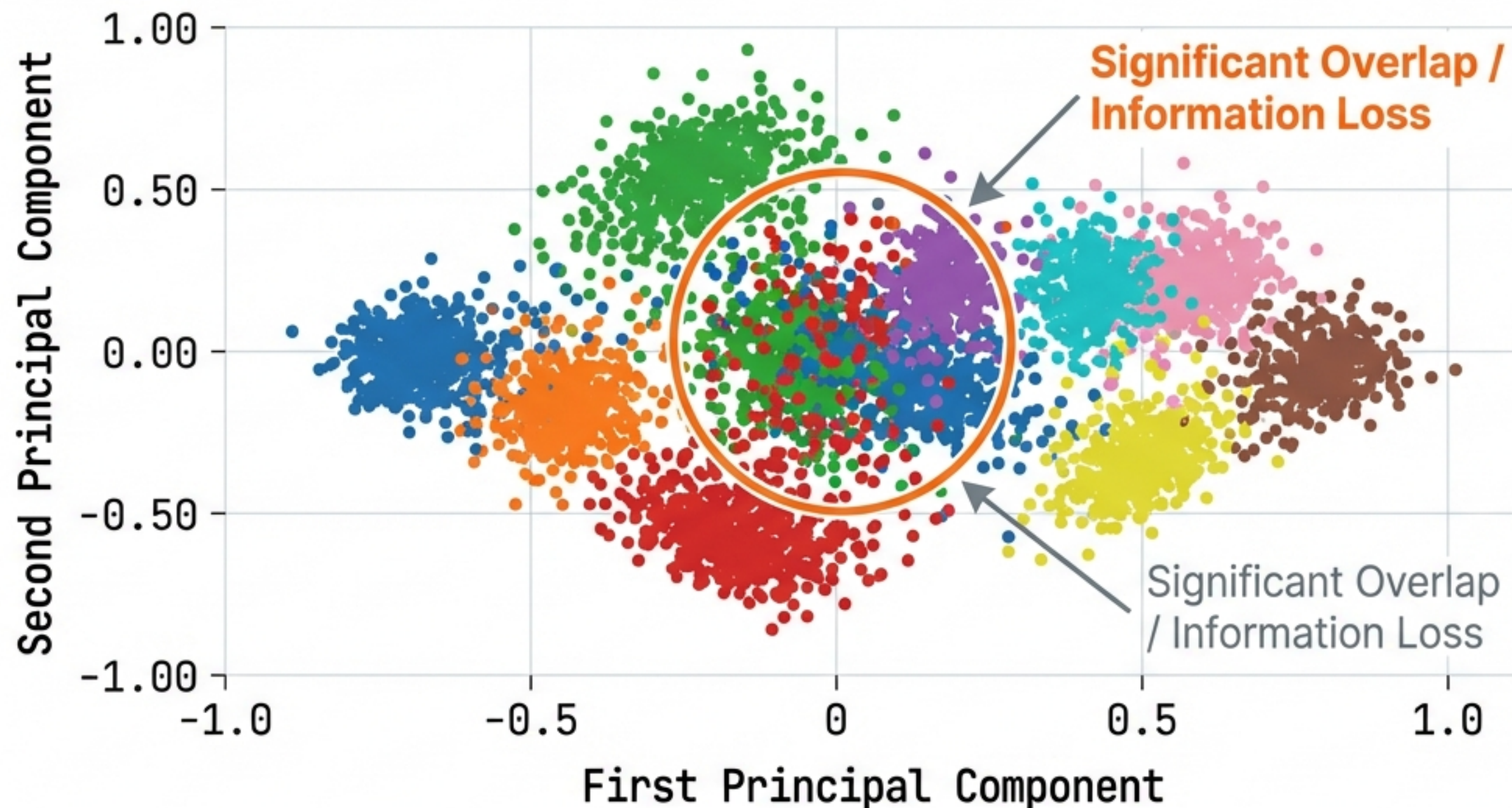
Beyond efficiency, PCA solves the visualisation problem. Humans cannot conceive of 784 axes, but we can interpret 2 or 3. By projecting the dataset onto its most significant eigenvectors, we can create human-readable maps of high-dimensional space.



Projection: Collapsing complexity into clarity.

The 2D Projection (PC1 vs PC2)

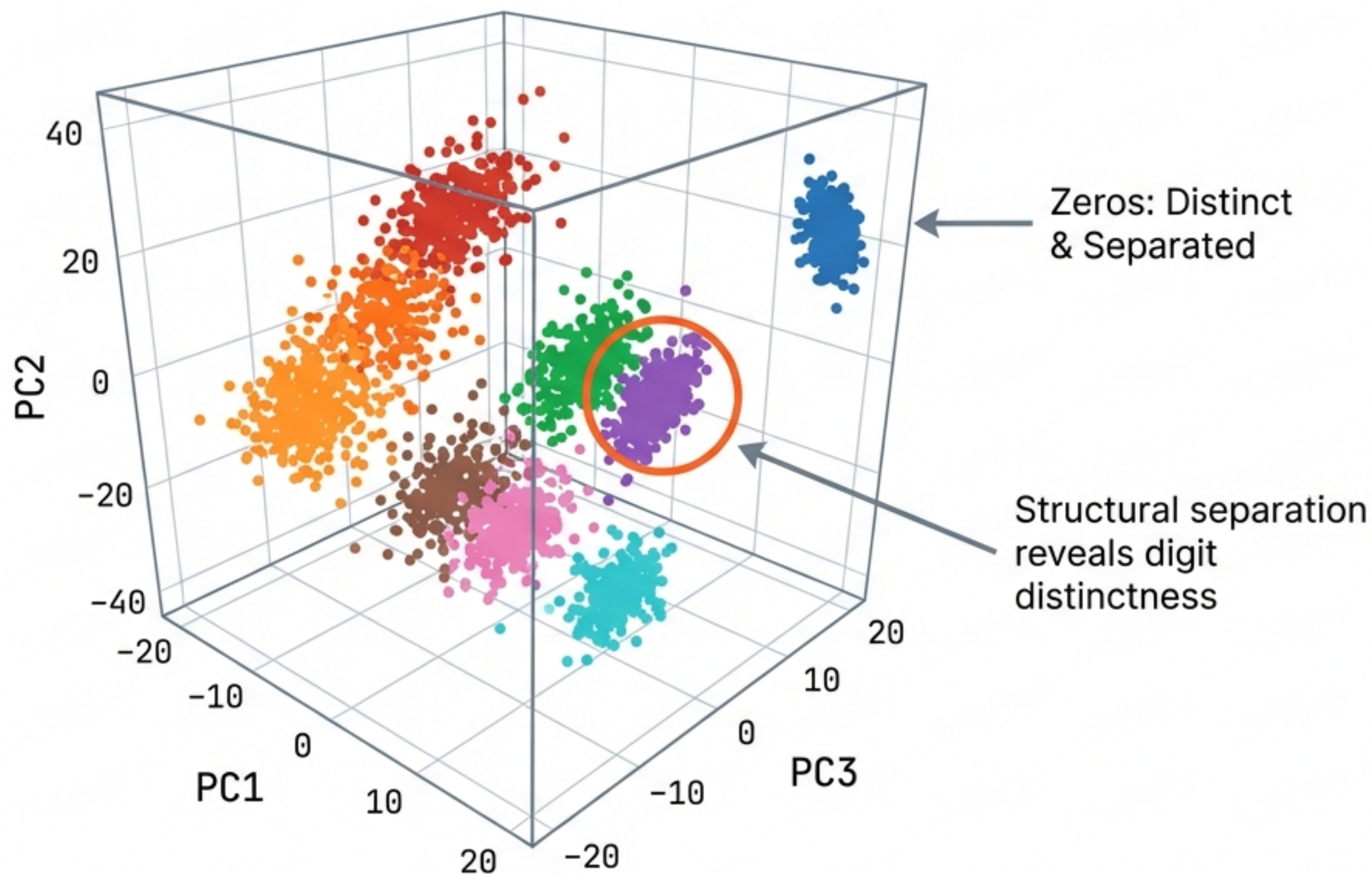
Projecting 784 dimensions onto a flat plane.



While broad clusters appear, nuance is lost. Digits hide behind one another in the flattened view.

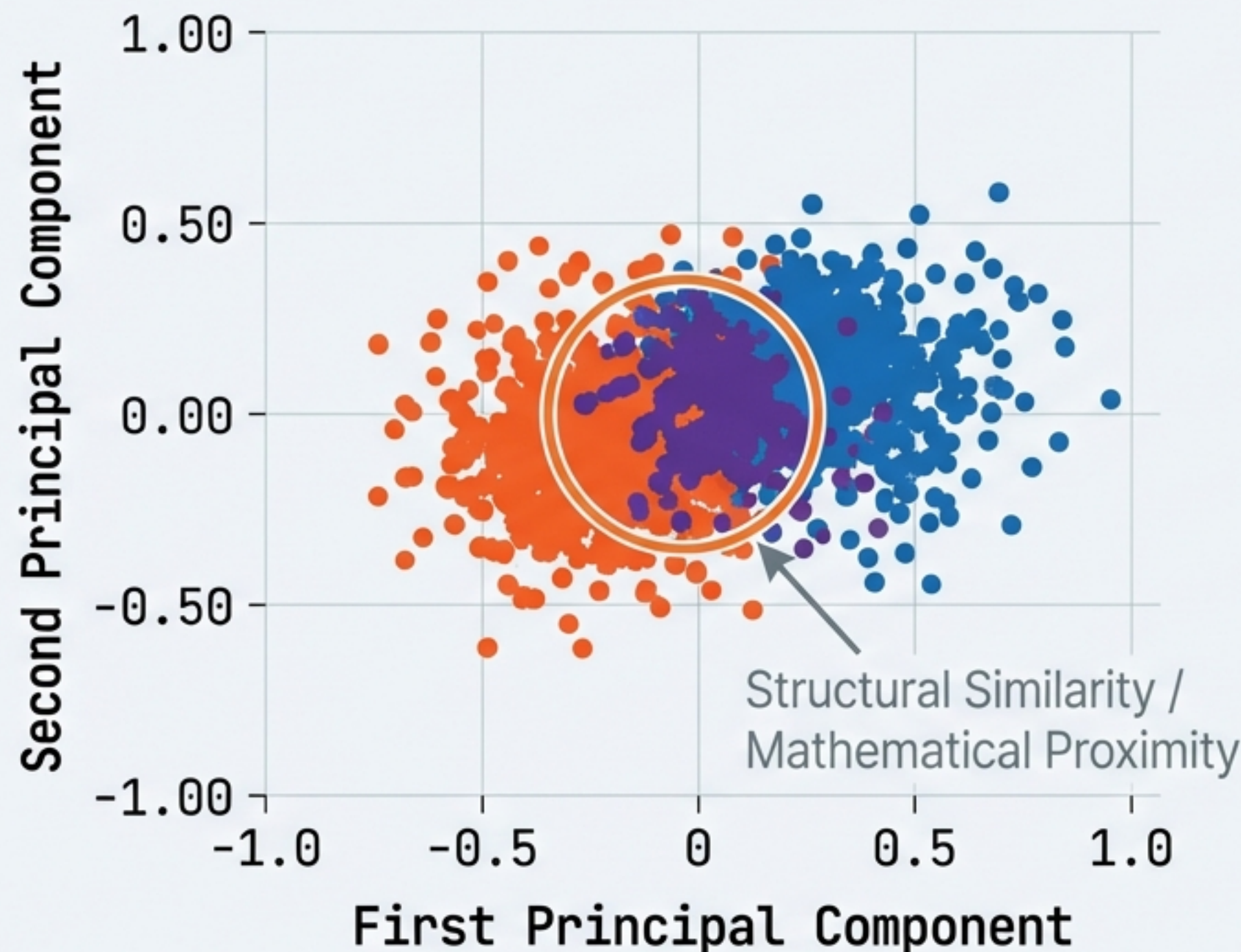
The 3D Projection: The Clarity of Depth

Adding the 3rd Component separates the clusters.



Interpreting the Visuals: Why Digits Overlap

Mathematical proximity reflects structural similarity.



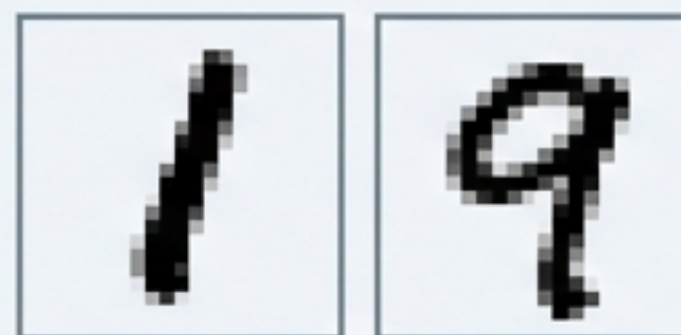
Visual Evidence



Handwritten digits, Slate Grey, and bitmps. Inter Light.



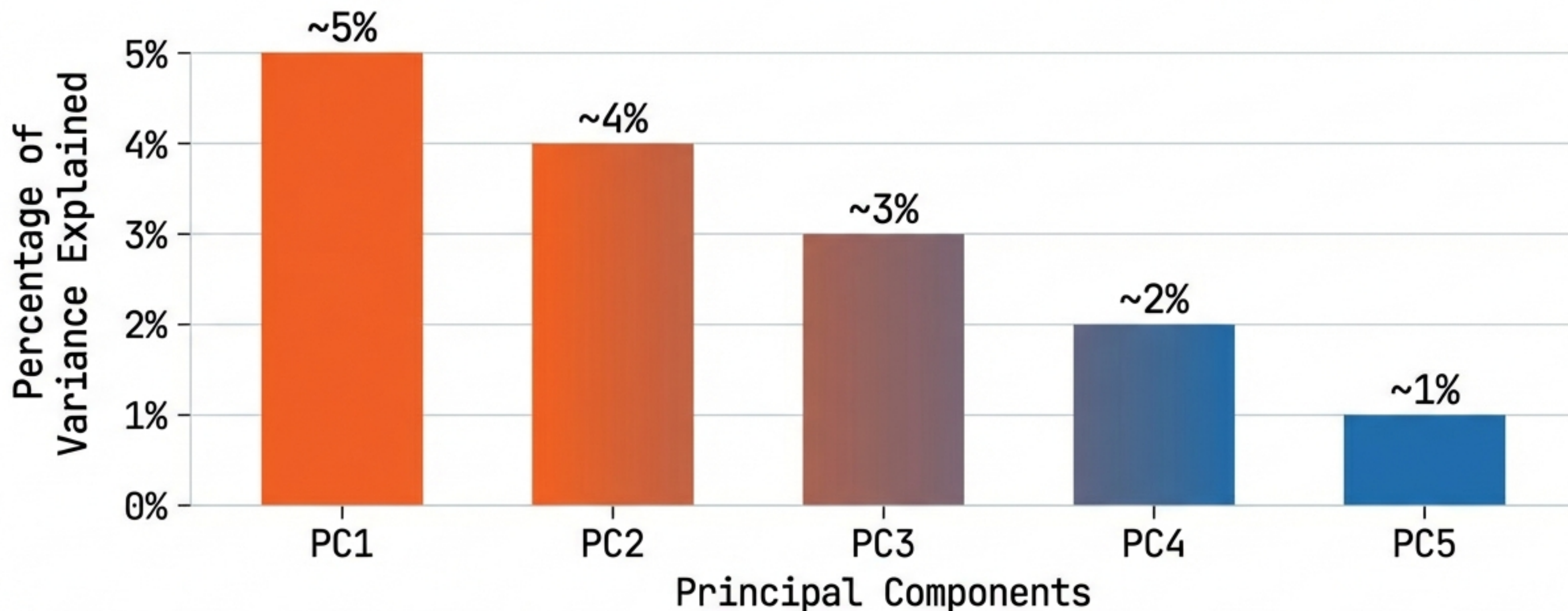
A '3' is structurally a subset of an '8'. To PCA, these vectors are close in space.



Shared vertical strokes create correlation between '1' and '9'.

Under the Hood: Eigenvalues & Variance

We select components based on `explained_variance_ratio_`. This metric tells us how much information each eigenvector holds.



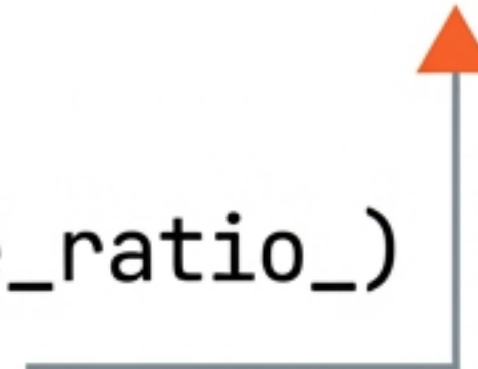
The 'power' of components diminishes rapidly. The first few hold the most structural information.

The Cumulative Sum Strategy

How to scientifically determine “n_components”.

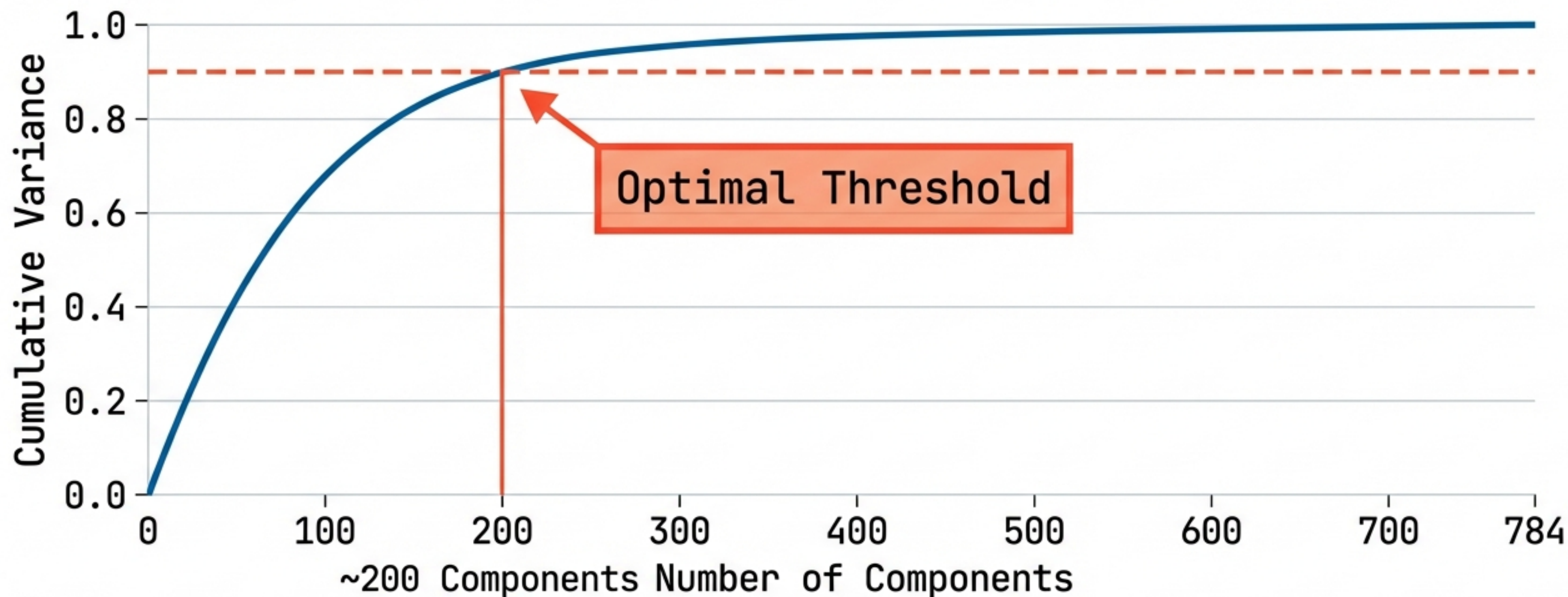
Target Variance $\geq$ 90%

```
import numpy as np
cumulative_variance = np.cumsum(pca.explained_variance_ratio_)
target_n = np.argmax(cumulative_variance >= 0.90) + 1
```



Instead of guessing, we calculate the cumulative sum of variance ratios. We retain enough components to preserve 90% of the original signal.

Finding the Optimal 'k': The Elbow Plot

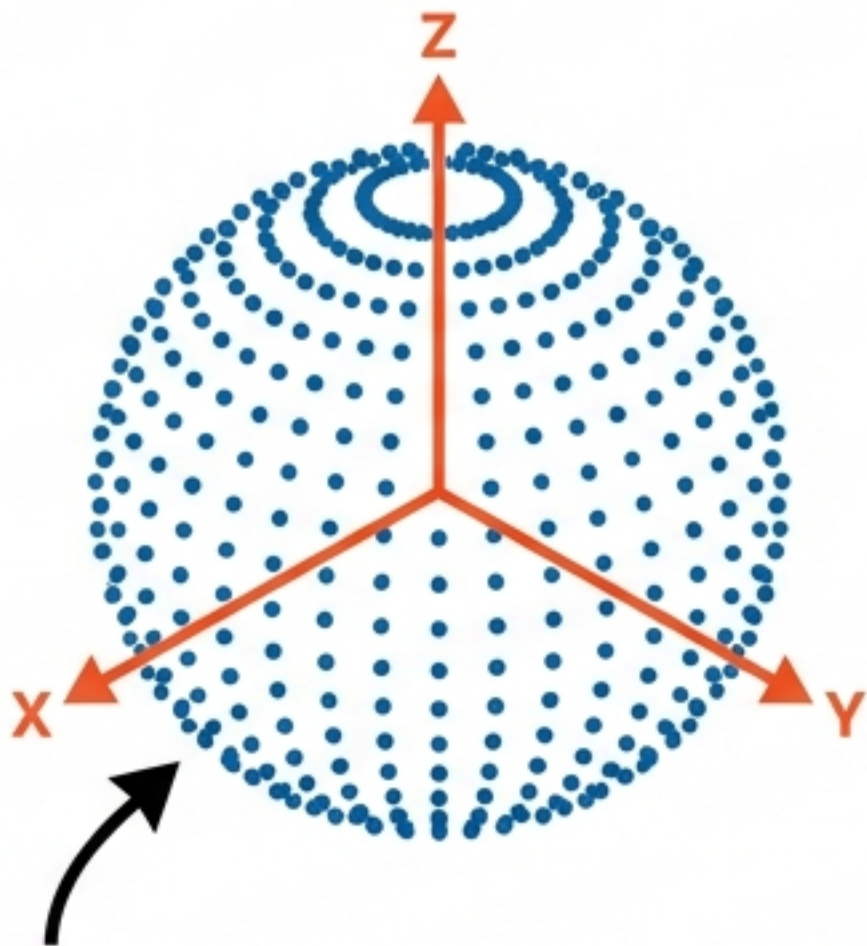


To preserve 90% of the MNIST dataset's information, we need approx. 200 dimensions. This reduces data size by 75%.

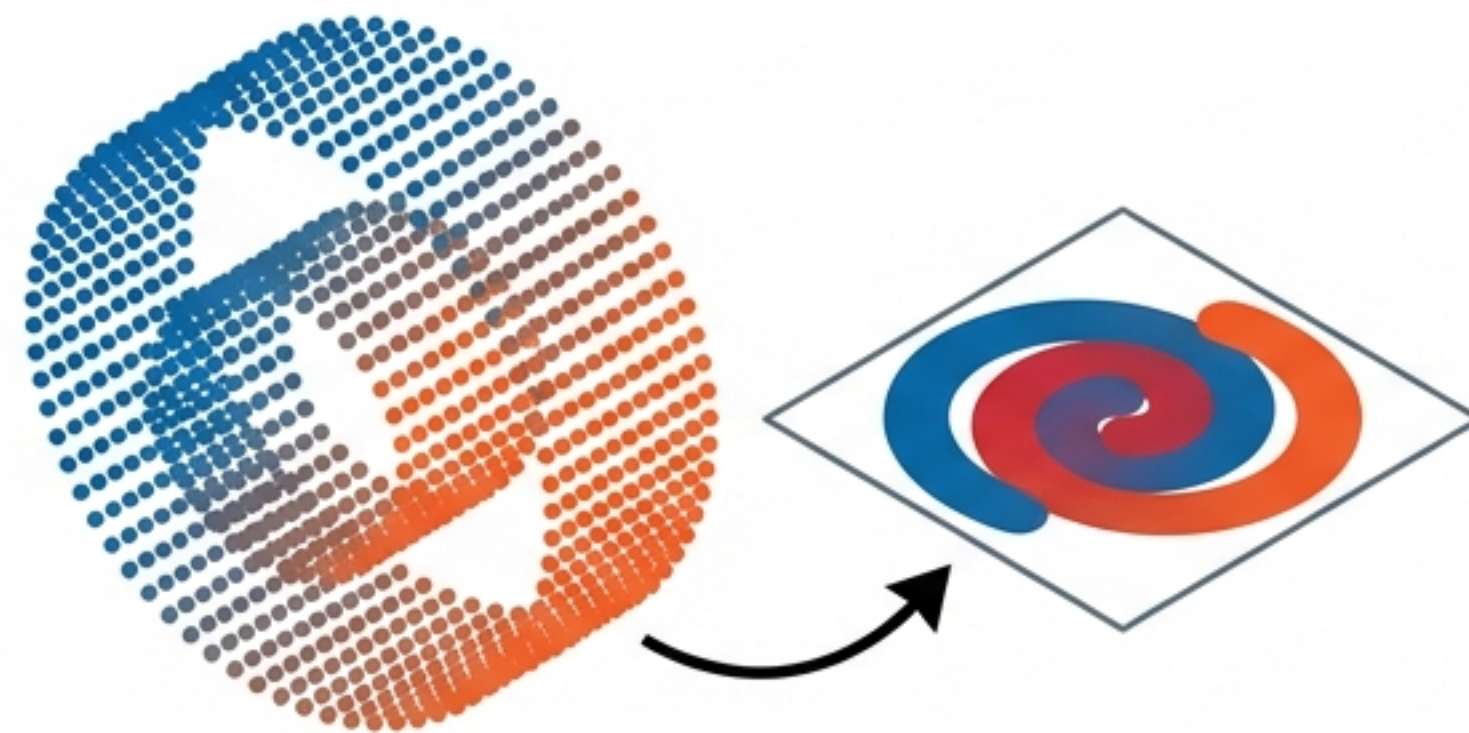
When PCA Fails

PCA assumes linear correlations. It struggles with equal variance or non-linear manifolds.

Failure Modes



Equal Variance = No Principal Direction.



Linear projection destroys non-linear structure.

Summary & Key Takeaways



Efficiency

Drastically reduced training time with minimal accuracy loss (**96.6% → 95.5%**).



Visualisation

Enables projection of N-dimensional data into 2D/3D space to reveal clusters and outliers.



The Trade-off

Use Cumulative Sum to balance Dimensions vs. Information. Aim for **90%** explained variance.

PCA transforms the 'Curse of Dimensionality' into manageable insights.